

STUDENT PROJECT TEMPLATE (SRS)

(INSTRUCTION FOR STUDENTS: THIS DOCUMENT SERVES AS THE FORMAL 'CONTRACT' FOR YOUR TEAM PROJECT. IT MUST BE MAINTAINED IN YOUR GITHUB REPOSITORY.)

POLYHOUSING

SOFTWARE REQUIREMENTS SPECIFICATION

INSTRUCTOR: LARA NICHOLS-BROWN

TERM: FALL 2026

SECTION: 03

THE ENGINEERING TEAM:

1. **STUDENT A:** AGUSTIN — SCRUM MASTER
2. **STUDENT B:** MIGUEL — PRODUCT OWNER
3. **STUDENT C:** FABIAN — LEAD ENGINEER

PROJECT ASSETS:

- **GITHUB:** [HTTPS://GITHUB.COM/AGUSTINDIMAYUGA/POLY-HOUSING](https://github.com/agustindimayuga/poly-housing)
- **DEPLOYMENT:** [LINK TO LIVE SITE]

DOCUMENT HISTORY:

LAST DATE CHANGED:	WHO:	WHAT WAS CHANGED:
WED SEPTEMBER 23, 2026	AGUSTIN XOCUA DIMAYUGA. FABIAN BALLESTEROS-LIMON, MIGUEL MEDINA	ADD USER STORIES, FUNCTIONAL REQUIREMENTS, NON- FUNCTIONAL REQUIREMENTS, AND PROJECT VISION + PROBLEM STATEMENT

**POLYHOUSING
SOFTWARE REQUIREMENTS SPECIFICATION (SRS)**

COURSE: CSC 3300

DATE: [SUBMISSION DATE]

TABLE OF CONTENTS

1. <u>INTRODUCTION</u>	<u>[PAGE #]</u>
○ <u>1.1 PROJECT PURPOSE</u>	
○ <u>1.2 INTENDED AUDIENCE</u>	
○ <u>1.3 PROJECT SCOPE</u>	
2. <u>USER STOREIES</u>	<u>[PAGE #]</u>
○ <u>2.1 USER FEATURES (THE "SHALL" STATEMENTS)</u>	
○ <u>2.2 ADMIN FEATURES</u>	
3. <u>FUNCTIONAL REQUIREMENTS</u>	<u>[PAGE #]</u>
4. <u>NON-FUNCTIONAL REQUIREMENTS</u>	<u>[PAGE #]</u>
○ <u>4.1 DATA INTEGRITY & SECURITY</u>	
○ <u>4.2 PERFORMANCE & USABILITY</u>	
5. <u>SYSTEM ARCHITECTURE</u>	<u>[PAGE #]</u>
○ <u>5.1 REST API ENDPOINTS</u>	
○ <u>5.2 DATABASE SCHEMA (MYSQL)</u>	
6. <u>USER INTERFACE (UI)</u>	<u>[PAGE #]</u>
○ <u>6.1 WIREFRAMES / MOCKUPS</u>	
7. <u>DATA REQUIREMENTS</u>	<u>[PAGE #]</u>
○ <u>7.1 LIST OF PERSISTENT DATA</u>	
8. <u>TRACEABILITY MATRIX</u>	<u>[PAGE #]</u>
9. <u>APPENDICES</u>	<u>[PAGE #]</u>

1. INTRODUCTION

- **PROBLEM STATEMENT: WHAT SPECIFIC PROBLEM ARE YOU SOLVING?**

Polyhousing is a Zillow alternative tailored for Cal Poly students. As students looking for off campus housing last year, we suffered through the struggles of using Zillow. Realtors not willing to rent to students, lack of accessibility, and landlords looking to fill vacancy immediately. Polyhousing proposes a single web application for all off-campus housing needs. All listings would be open to students; students are able to sub-lease on a single platform rather than using social media apps, and students are guaranteed to not struggle through the process of finding a place to live.

- **TARGET AUDIENCE: WHO ARE THE STAKEHOLDERS?**

- **SCOPE: WHAT IS THE SCOPE OF THIS PROJECT?**

- **PROJECT VISION**

For Cal Poly students who struggle to find accessible off-campus housing and sublease opportunities, PolyHousing is a student-focused housing platform that provides a centralized place to search for housing, find subleases, and connect with landlords and other students. Unlike Zillow and social media platforms, PolyHousing is designed specifically around the needs of Cal Poly students, bringing student-friendly housing listings and subleases together in one platform. - Agustin Xocua Dimayuga

2. USER STORIES

USER STORIES FOLLOW THE FORMAT: "AS A [TYPE OF USER], I WANT TO [ACTION] SO THAT [VALUE/BENEFIT]."...

- US-01: AUTHENTICATION
- US-02: STORE WORKOUT LOG.
- US-03: SEARCH HISTORY

ID	REQUIREMENT	PRIORITY
US-01	As a student, I want to be able to look for the realtor's contact info so that I can learn more about a listing. - Fabian Ballesteros-Limon	2
US-02	As a student, I want to be able to create an account and log in so that I can start looking for houses. - Fabian Ballesteros-Limon	1
US-03	As a homeowner/realtor, I want to be able to post a listing for a house so that students can look and apply for it. - Fabian Ballesteros-Limon	3
US-04	As a student, I want to save a house so that I can look back on it on the future - Agustin Xocua	4

US-05	As a student, I want to view detailed information about housing listings so that I can determine whether the property is suitable for me. - Agustin Xocua	5
US-06	As a student, I want to post a sublease listing so that I can find another student to take over my lease over the summer. - Agustin Xocua	6
US-07	As a student I want to leave a review on a house based on my experience living there so that other students have an insight on what the living situation might be - Miguel Medina	7
US-08	As a homeowner I want to input what utilities and extra expenses will be covered so that students know ahead of time - Miguel Medina	8
US-09	As a student I want to view the houses for a specific realtor company so that I can reuse their application - Miguel Medina	9

3. FUNCTIONAL REQUIREMENTS

THE SYSTEM SHALL...

ID	REQUIREMENT	PRIORITY
FR-01	The system shall allow users to create an account, authenticate using their username and password, and log out. - Agustin Xocua Dimayuga	1
FR-02	The system shall allow users to search and filter available housing listings based on criteria such as price, location, bedrooms, housing type, and availability. - Agustin Xocua Dimayuga	1
FR-03	The system shall allow users to view detailed information about a housing listing, including its location, price, availability, housing type, and description. - Agustin Xocua Dimayuga	1
FR-04	The system shall allow students, landlords, and property managers to create housing listings, including sublease listings. - Agustin Xocua Dimayuga	11
FR-05	The system shall allow listing owners to edit, update the availability of, and delete their listings. - Agustin Xocua Dimayuga	1

FR-06	The system shall allow users to save housing listings for later viewing. -Agustin Xocua Dimayuga	2
FR-07	The system shall allow users to contact the owner of a housing listing through the application. -Agustin Xocua Dimayuga	1
FR-08	The system shall allow users to report listings that are inaccurate, inappropriate, or potentially fraudulent. - Agustin Xocua Dimayuga	2
FR-09	The system shall identify whether a listing is a standard rental or a sublease. - Agustin Xocua Dimayuga	2
FR-10	The system shall display when a listing was created or last updated. - Agustin Xocua Dimayuga	3

4. NON-FUNCTIONAL REQUIREMENTS

- **INTEGRITY: HANDLING ERRORS AND BAD DATA**
- **SECURITY: IMPLEMENT BCRYPT FOR PASSWORD HASHING AND ENVIRONMENT VARIABLES FOR DB CREDENTIALS.**
- **USABILITY:**

<u>ID</u>	<u>REQUIREMENT</u>	<u>PRIORITY</u>
<u>NON-FR-1</u>	Product should be able to verify a house owner from anyone else -Miguel Medina	
<u>NON-FR-2</u>	Product should be able to handle many users during specific months (Oct -> Dec) & (April -> June) -Miguel Medina	
<u>NON-FR-3</u>	Product should only show houses that are actively looking for tenet -Miguel Medina	
<u>NON-FR-4</u>	Product should securely save private information without directly showing it in the DB. -Miguel Medina	
<u>NON-FR-5</u>	Product should properly and correctly identify a user based on a unique id -Miguel Medina	

5. SYSTEM ARCHITECTURE

- **REST ENDPOINTS: (LIST METHOD | URL | DESCRIPTION)**
- **LIST YOUR PLANNED ENDPOINTS BASED ON YOUR REST SLIDE:**
 - **GET /API/RESOURCES — FETCHES COLLECTION.**
 - **POST /API/RESOURCES — CREATES A NEW RESOURCE.**
- **UML: (ATTACH A DIAGRAM)**

5. USER INTERFACE (UI)

A. 5.1 WIREFRAMES / MOCKUPS

6. DATA REQUIREMENTS

7. TRACEABILITY MATRIX

ID	REQUIREMENT	LINE OF CODE
US-01		
US-02		
US-03		
US-04		

8. AI USAGE & DISCLOSURE (MANDATORY)

- **MODEL(S) USED: [E.G., CLAUDE 3.5, GPT-4O]**
- **PROMPTS USED DURING CODING:**